Predicting NVIDIA's Next-Day Returns and Volatility: An Empirical Analysis Using ARIMA and GARCH Models

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Abstract

This project evaluates the ability of the ARIMA and GARCH models to forecast the daily returns and volatility of NVIDIA's shares. We used daily adjusted closing prices from March 2020 to March 2025, transformed into log-returns to achieve stationarity. For return forecasting, various ARIMA models are estimated and compared using an expanding window one-step-ahead forecast approach with RMSE and MAE as accuracy metrics. The ARIMA(1,0,0) model performs best in this framework, but its forecasts are only marginally better than a naïve benchmark due to the high randomness of the stock returns. To forecast volatility, a GARCH(1,1) model is fitted to the ARIMA residuals, capturing the well-known clustering in volatility. The out-of-sample forecasts from the GARCH model align reasonably well with the actual volatility measures. These findings are aligned with financial theory and literature: short-term returns remain largely unpredictable, whereas volatility shows enough persistence to allow for modest predictability.

Keywords: NVIDIA, ARIMA, GARCH, return forecast, volatility forecast, time series analysis, stock returns, conditional heteroskedasticity, model evaluation, expanding window forecast, financial econometrics

1 Introduction

Forecasting plays a central role in financial markets as investors, analysts, and risk managers continuously seek to predict future returns or changes in volatility. This study focuses on NVIDIA Corporation, a leading technology firm in the semiconductor and artificial intelligence sectors, as a case for forecasting stock behavior. NVIDIA's stock is of interest due to its role in a rapidly evolving industry and its appeal to institutional and retail investors. This broad interest and the substantial price swings of the stock make NVIDIA an interesting topic to examine the dynamics of return and volatility.

Financial theory and empirical evidence suggest that stock returns are difficult to forecast, often behaving like random noise, while volatility tends to exhibit a more prominent structure (Fama, 1970). Periods of high and low volatility often cluster together, a phenomenon known as volatility clustering. Because of these properties, ARIMA models are applied to forecast the returns (mean of the series), and GARCH models are used for time-varying volatility. We apply these models to NVIDIA to evaluate their forecast performance in practice.

ARIMA and GARCH have been used in multiple studies before, including individual stocks such as NVIDIA (Albta Zková, Jitka Veselá, 2023). What sets this project apart is not the models themselves, but our dynamic, expanding-window evaluation strategy. The analysis focuses on one-step-ahead forecasts, using an expanding window of past data to predict the next day's return or volatility. This approach is meant to reflect a realistic forecasting setting, where the model is updated over time and only relies on information available at each point. We evaluate model performance over a designated out-of-sample period to assess how well the models perform in practice.

The data consist of daily adjusted closing prices of NVIDIA from March 2020 to March 2025. We transform the price series into logarithmic returns to stabilize the variance and satisfy stationarity requirements. An augmented Dickey-Fuller test confirms that the log-return series is stationary, justifying the use of ARIMA models without additional differencing. Several ARIMA model specifications are then fitted to the return series. The competing models are compared using the information criteria BIC and out-of-sample forecast accuracy, the best ARIMA model is selected as the mean model. Using the residuals from this ARIMA model, we subsequently estimate a GARCH model to capture conditional heteroskedasticity. Finally, we generate one-step-ahead forecasts for both returns and volatility and evaluate their accuracy.

In summary, the project addresses the question: *Can ARIMA and GARCH models provide useful forecasts for NVIDIA's daily returns and volatility?* To preview the findings: while ARIMA shows limited effectiveness in forecasting returns, the GARCH model performs notably better when modeling and predicting volatility.

2 Data

This project uses daily stock price data for NVIDIA Corporation, extracted from Yahoo Finance. The sample ranges from March 3, 2020, to February 28, 2025, providing a five-year window that includes both calm and volatile markets. This period begins shortly before the COVID-19 pandemic, an event that triggered significant financial volatility, and extends through recent years marked by elevated interest rates and rapid expansion in the technology sector. Daily frequency was chosen to capture short-term dynamics, as both ARIMA and GARCH rely on detecting patterns that only appear in day-to-day price movements. The choice of start and end dates also reflect data availability, with March 3, 2020, marking the earliest trading day, and February 28, 2025, chosen as the final trading day.

Figure 1: NVIDIA Stock Prices

The prices used are the adjusted closing prices, which account for dividends and stock splits, allowing for a cleaner return calculation. The data are recorded at a daily frequency, meaning weekends and market holidays are excluded. No seasonal adjustments were made, and the dataset contains no missing observations. All dates were checked to ensure continuity in trading days, and no gaps or irregularities were found during the data cleaning process.

2.1 Data Split

To evaluate our forecasting models properly, the NVIDIA stock data were portioned into three subsets. Specifically, we divided the sample into a training set, a validation set and a test set, corresponding to the earliest 60%, the next 20%, and the final 20% of the observations, respectively. The training set is used to fit and estimate model parameters. The validation set is held out from the model fitting and is used to compare candidate model forecasts. The test set is reserved entirely for the final evaluation of the chosen models.

For the ARIMA analysis, this three-way split ensures that hyperparameters (such as the lag orders) are selected based on out-of-sample performance on the validation set, without peeking at the test data. In contrast, the GARCH analysis is treated with a simpler two-way split: we fit only on the training data and then evaluate its forecasts on the test set. Since we only evaluate a single GARCH model specification, there is no need for a separate validation set. This setup ensures that no information from the test period influences the estimation or specification of the GARCH model. The figure below visually illustrates this structured data partitioning:

Figure 2: Train, validation, and test split of NVIDIA stock prices

2.2 Stationarity

Modelling time series data requires the series to be stationary, meaning its mean, variance and autocovariance remain constant over time. Non-stationary series can lead to spurious results in ARIMA and GARCH-type models, so stationarity must be assessed and ensured before proceeding.

To stabilize the variance, we first apply a log transformation to the price series. Visual inspection of the log-transformed series still reveals an upward trend, indicating potential non-stationarity in the mean. To formally assess this, we conduct the Augmented Dickey-Fuller (ADF) test on both the raw and log-transformed series. In both cases, the test statistic exceeds the critical values, meaning we fail to reject the null hypothesis of a unit root, confirming non-stationarity:

To address this, the log-transformed series is differenced once to compute daily logarithmic returns, defined as:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right) \tag{1}$$

The ADF test is then applied again to the return series. This time, the test statistic is well below the critical values, strongly rejecting the null hypothesis and confirming that the return series is stationary in mean.

Test Statistic	**Critical Values**		
	1%	5%	10%
-6.7963	-2.58	-1.95	-1.62

Table 1: ADF Test Results for Log-Return Series

In addition to stationarity in the mean, volatility in the return series appears time-varying as seen in Figure 3, which is a common feature in financial data. This conditional heteroskedasticity is not addressed by ARIMA models, which assume constant variance. To model this behavior, we will later turn to the GARCH model. This two-step approach allows for separate modeling of the conditional mean and variance.

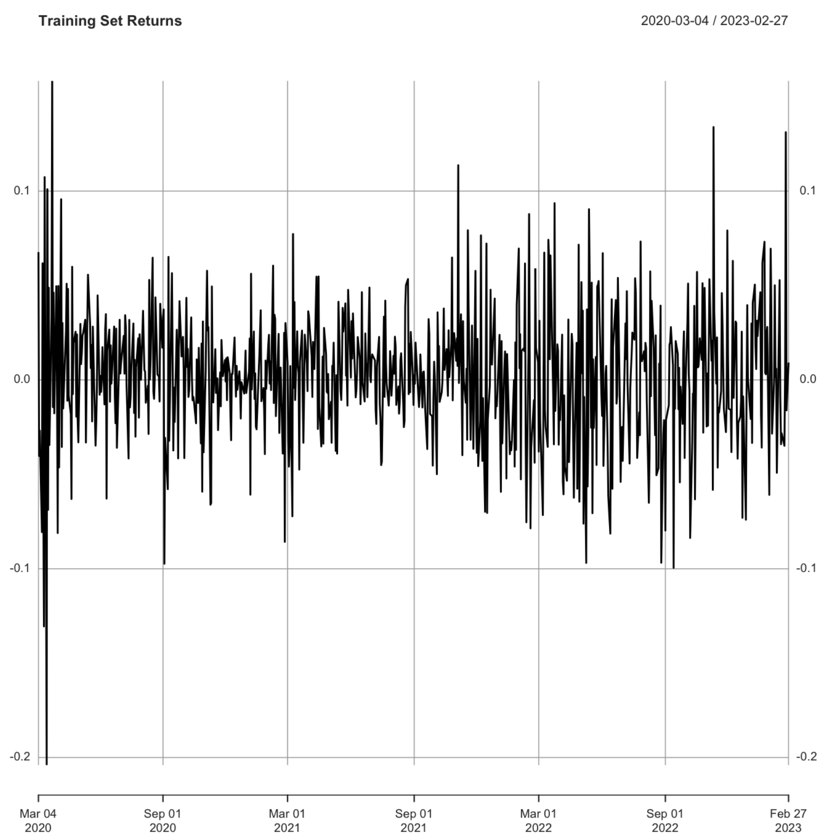


Figure 3: Returns in Training Set

3 Methods

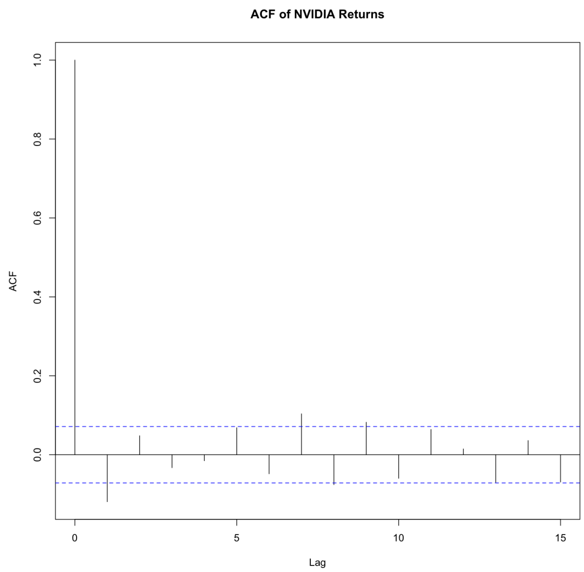
3.1 ARIMA Model for Returns

ARIMA is short for *Autoregressive Integrated Moving Average*, and incorporates past values (AR), differencing for stationarity (I), and past errors (MA). In an ARIMA(p, d, q) model, the series is differenced d times to achieve stationarity, then modeled as an ARMA(p, q) process. In our case, since log-returns are already stationary, we consider ARIMA$(p, 0, q)$ models, which reduce to ARMA(p, q).

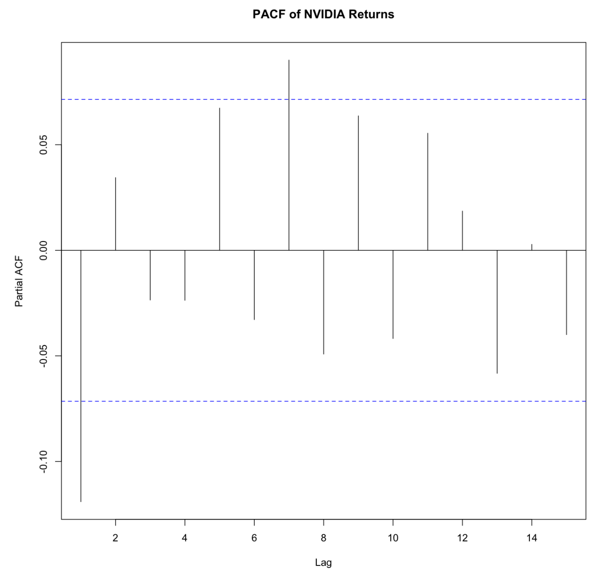
In its general form, an ARIMA model is written as:

$$y_t = a_0 + \sum_{i=1}^{p} a_i y_{t-i} + \sum_{i=0}^{q} \beta_i \varepsilon_{t-i} \tag{2}$$

Where a_0 is a constant term, a_i are the *autoregressive* coefficients (AR), β_i are the *moving average* coefficients (MA), and ε_t is white noise, the unexpected part of the return.



(a) ACF of NVIDIA Returns



(b) PACF of NVIDIA Returns

Figure 4: ACF and PACF plots of NVIDIA's daily log-returns.

These plots of the autocorrelation and partial autocorrelation show minor autocorrelation at short lags, suggesting that there is still some structure in the mean that can be modeled. ARIMA models are designed to capture this kind of short-term dependence of previous values by combining past values and past residuals. Based on PACF, which shows a significant spike at lag 1, an AR(1) process may be appropriate. Meanwhile, the ACF shows a quick drop after lag 1, consistent with a possible MA(1) component. These indications suggest that ARIMA(1,0,0) or ARIMA(1,0,1) may be potential candidates for forecasting.

Another reason for supporting our choice of ARIMA is that it is widely used and well supported in statistical software and fits naturally within the structure of our data.

3.2 GARCH Model for Volatility

After fitting the ARIMA model to the return series, we took a closer look at the residuals. While the ARIMA model captured the patterns in the mean, a plot of the squared residuals revealed something interesting: the variance of these errors was not constant. Some periods showed small fluctuations, while others had much larger spikes in the residual size.

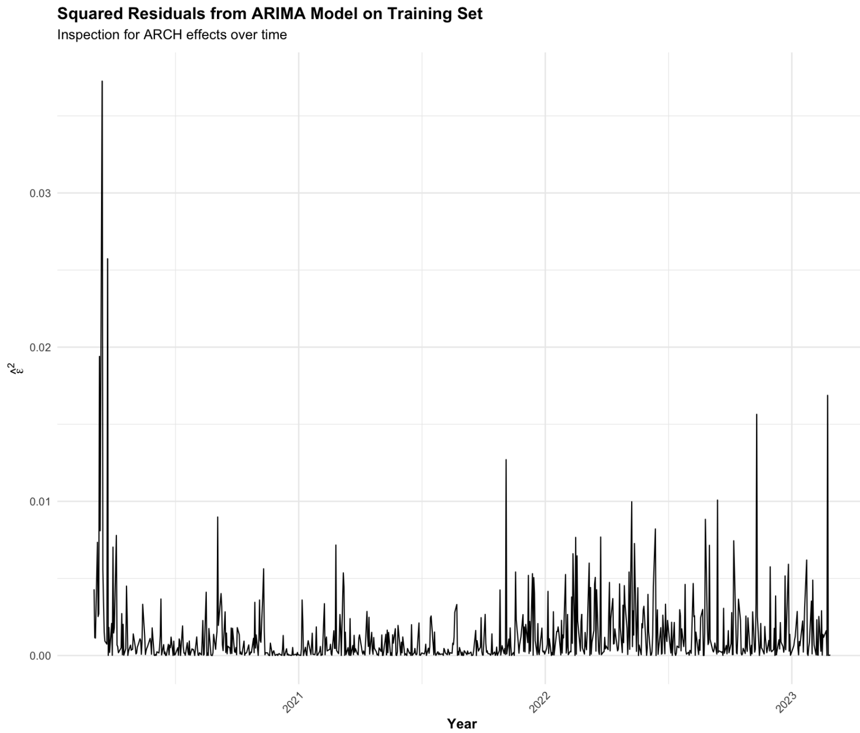


Figure 5: Squared Residuals from ARIMA(1,0,0) Model on Training Set

This behavior, where periods of high volatility tend to be followed by more volatility, and calm periods tend to cluster together, is known as *volatility clustering*, and it's a sign of *conditional heteroskedasticity.* In other words, the size of the part that the ARIMA model doesn't capture changes over time. To model this changing variance, we use a GARCH(p, q) model, which stands for *Generalized Autoregressive Conditional Heteroskedasticity.*

GARCH builds on the ARCH framework by including past variances with past squared return shocks. In the general form, the conditional variance is specified as:

$$\sigma_t^2 = \omega + \sum_{i=1}^{q} \alpha_i \varepsilon_{t-i}^2 + \sum_{i=1}^{p} \beta_i \sigma_{t-i}^2 \tag{3}$$

This formulation implies that today's volatility depends on both past squared shocks ε_{t-i}^2 and past volatility σ_{t-i}^2, with parameters $\omega > 0$, $\alpha_i \geq 0$ and $\beta_i \geq 0$ ensuring a well-defined variance process. In the terminology of Enders (Enders, 2014), this model is often rewritten in terms of the conditional variance h_t, giving:

$$\varepsilon_t = v_t \sqrt{h_t} = v_t \sqrt{\sum_{i=1}^{q} \alpha_i \varepsilon_{t-i}^2 + \sum_{i=1}^{p} \beta_i h_{t-i}} \tag{4}$$

Here, v_t is a random shock with zero mean and constant variance, while h_t shows how much volatility we expect at time t. This equation tells us that the size of the return shock of today depends on how volatile we expect the market to be. When volatility is high, shocks tend to be large, and when it is low, shocks are expected to be small. This helps the GARCH model capture volatility clustering, where high or low volatility persists. This setup works a lot like an ARIMA model, but instead of modeling the mean of the returns, it models how the variance changes over time. This general framework provides the basis for estimating and forecasting volatility in our time series.

3.3 Model Selection With BIC

To determine the most appropriate models for our forecasting task, we relied on the Bayesian Information Criterion (BIC), which penalizes model complexity while rewarding goodness of fit. This criterion helps prevent overfitting by discouraging unnecessarily complex models that do not significantly improve predictive performance. The BIC is defined as:

$$BIC = -2\ln(L) + k\ln(n) \tag{5}$$

where L is the model likelihood, k is the number of parameters, and n is the sample size. Lower BIC values indicate a better balance between fit and simplicity.

This model selection strategy ensures that ARIMA models are chosen based on their in-sample fit to the training data, as measured by the BIC. The models with the lowest BIC are then evaluated on their forecasting performance using the validation and test sets. For the GARCH model, we estimate parameters once using the full training data (80% of the data) and assess out-of-sample volatility forecasts without reselecting or re-tuning the model.

3.4 Out-of-sample Forecasting

To evaluate the models in a realistic setting, we design an out-of-sample forecasting exercise using a 60-day expanding window approach, which is roughly three trading months. We split the dataset into an initial training period, validation period, and a test period for forecasting evaluation. Then for each day in the validation and test period, we proceed as follows: (1) estimate the ARIMA model on the initial 60-day window, (2) forecast the next day's return, (3) use the ARIMA residuals up to the current day to estimate the GARCH(1,1) model and forecast the next day's volatility, and (4) expand the sample to include the actual next day value (once observed) and repeat. This means the model is re-trained continuously, growing the sample as new observations arrive, which mimics how a forecaster in real time would update their model with the latest information.

We chose a 60-day initial window somewhat arbitrarily, as our goal is demonstration of the models rather than fine tuning parameters. For evaluating volatility forecasts, we need a proxy for "realized" volatility on each day of the test set since the true volatility is unobservable. A 30-day rolling standard deviation of returns was used as a comparative benchmark for the true volatility. In other words, for each day in the test period we compute the standard deviation of returns over the previous 30 trading days as an empirical measure of the volatility. The one-step GARCH forecast is then compared to this realized volatility measure of the next day.

To evaluate the accuracy of the return and volatility forecasts, we use two common error metrics: Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE). As the forecasts and actual values are daily log returns in percentage terms, the RMSE and MAE metrics reflect average forecast errors in percentage points. These measure the average size of the forecast errors, where y_i is the actual value, y_p is the predicted value, and n is the number of forecast observations. Mathematically, they are defined as:

$$RMSE = \sqrt{\frac{1}{n}\sum_i (y_i - y_p)^2}, \quad MAE = \frac{1}{n}\sum_i |y_i - y_p| \tag{6}$$

These metrics allow us to quantify how closely the ARIMA and GARCH models' forecasts align with the realized return and volatility.

3.5 Model Diagnostics with Ljung-Box Test

We assess the adequacy of our models using the Ljung–Box test to detect any residual autocorrelation, which would suggest model misspecification. For each candidate ARIMA model, we apply this test to both the raw residuals and their squared values. If the ARIMA model sufficiently captures the mean dynamics, the raw residuals should resemble white noise. However, if volatility clustering remains unaccounted for, the squared residuals may still show significant autocorrelation, indicating the presence of ARCH effects.

In our results, the raw residuals of each ARIMA model show no significant autocorrelation (Ljung–Box p-values consistently > 0.05 for up to 30 lags), confirming proper mean specification. However, the squared residuals display significant autocorrelation, revealing conditional heteroskedasticity in NVIDIA's returns and justifying the need for a GARCH specification. To capture this time-varying volatility structure, we fit a GARCH(1,1) model to the ARIMA residuals and reapply the Ljung–Box test, this time to the standardized residuals. The GARCH model is considered well-specified if these standardized residuals, and their squares, show no significant autocorrelation. Across our fitted models, the null hypothesis of white noise is not rejected, suggesting that our models successfully capture the underlying structure in both the mean and variance of the returns.

4 Results

4.1 ARIMA Model Estimation on Training set

Five candidate ARIMA(p,0,q) models were estimated using the training sample of daily log-returns. The models selected for evaluation, ARIMA(1,0,0), ARIMA(2,0,0), ARIMA(0,0,1), and ARIMA(0,0,2), were those with the lowest Bayesian Information Criterion (BIC) values, with ARIMA(2,0,1) additionally identified by the auto.arima() function as a strong candidate. Among these, ARIMA(1,0,0) achieved the lowest overall BIC (≈ -2867.55), indicating the best trade-off between fit and complexity.

Model	Coefficients (s.e.)	MAE	RMSE	BIC
ARIMA(1,0,0)	ar1 = -0.1195 (0.0363)	0.02669	0.03580	-2867.55
ARIMA(2,0,0)	ar1 = -0.1154 (0.0365), ar2 = 0.0348 (0.0365)	0.02667	0.03546	-2861.84
ARIMA(0,0,1)	ma1 = -0.1114 (0.0350)	0.02669	0.03570	-2866.78
ARIMA(0,0,2)	ma1 = -0.1131 (0.0365), ma2 = 0.0411 (0.0362)	0.02668	0.03547	-2861.44
ARIMA(2,0,1)	ar1 = 0.7149 (0.2139), ar2 = 0.1157 (0.0394), ma1 = -0.8332 (0.2123)	0.02672	0.03547	-2854.93

Table 2: ARIMA Model Estimates and Training Set Forecast Accuracy

Forecast error measures on the training set also were remarkably similar across all models, with RMSE values clustered around 0.0355 and MAE values near 0.0266. This similarity reflects the fact that our daily return series exhibit little serial dependence and are often close to white noise. As such, even models with additional AR or MA terms offer minimal gain in capturing structure, and their forecasts do not meaningfully improve upon those from simpler specifications. The ARIMA(1,0,0) model included a statistically significant AR(1) coefficient of -0.1195, indicating a mild mean reversion. In contrast, the additional parameters in more complex models were smaller and often statistically insignificant. These findings suggest that ARIMA(1,0,0) is a strong candidate, but all models are carried forward to the validation phase to formally compare their out-of-sample forecasting performance before final selection.

4.2 ARIMA Forecasting Performance on Validation Set

The five models were evaluated on the validation set using an expanding window one-step-ahead forecasting procedure. Specifically, for each day of the validation period the model was re-estimated using all available past data, and a one-step-ahead return forecast was generated.

Model	RMSE	MAE
ARIMA(1,0,0)	0.0286	0.0202
ARIMA(0,0,1)	0.0286	0.0202
ARIMA(0,0,2)	0.0295	0.0206
ARIMA(2,0,0)	0.0292	0.0205
ARIMA(2,0,1)	0.0297	0.0209

Table 3: Validation Set Forecast Accuracy for ARIMA Models

Forecast accuracy was evaluated using *RMSE* and *MAE*. The differences between the models were minimal, with *ARIMA*(1,0,0) slightly outperforming the others (*RMSE*=0.0286, *MAE*=0.0202). These findings are consistent with the findings on the training set, namely, that the return series exhibits little serial dependence, limiting the forecastable structure.

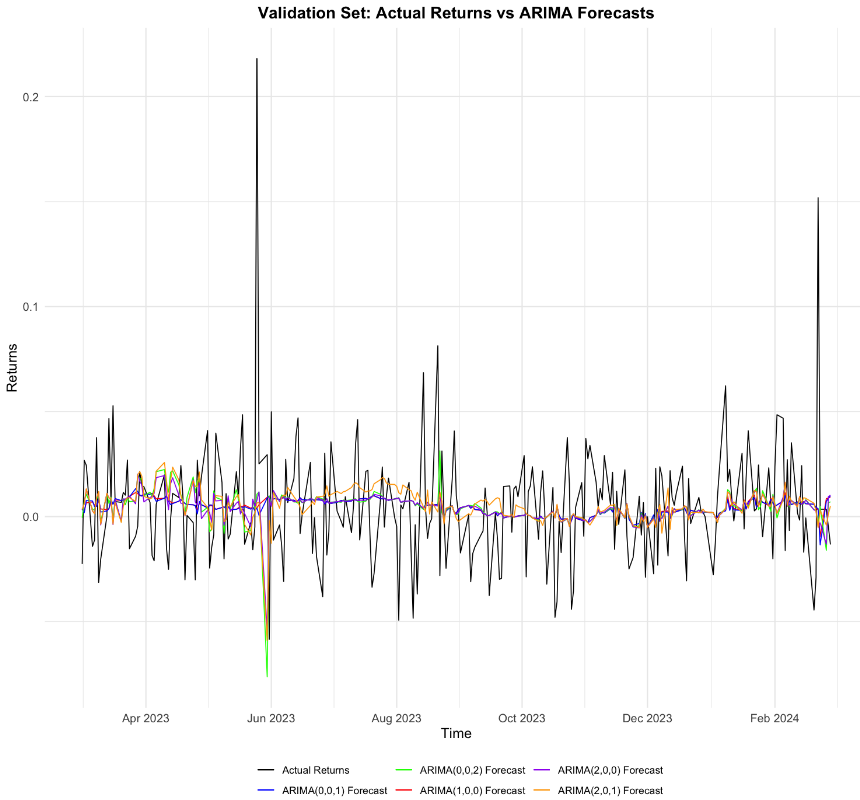


Figure 6: ARIMA Validation Set Forecasts

Figure 6 illustrates the actual returns alongside forecasts from all five models. All ARIMA forecasts remain close to the mean and cluster tightly together, confirming the statistical results: model complexity makes little difference in forecasting daily returns for NVIDIA. Despite the narrow performance gap, we retain **ARIMA(1,0,0)** as our final model based on its lowest errors and simplicity.

4.3 Final Forecast and Evaluation on Test Set

With ARIMA(1,0,0) selected as the best performing model on the validation set, the model was applied to the test set using the same procedure as before. On each test day, ARIMA(1,0,0) was re-estimated on all available prior data used to predict the next day's return. The forecasted values were compared to the actual observed test returns. The final performance metrics were **RMSE** = 0.0362 and **MAE** = 0.0271. These error values are only marginally worse than those observed during the validation period, indicating consistent predictive ability across both out-of-sample periods. The slightly elevated RMSE in the test set likely stems from occasional spikes in actual returns, which have a larger impact on RMSE due to its sensitivity to large errors.

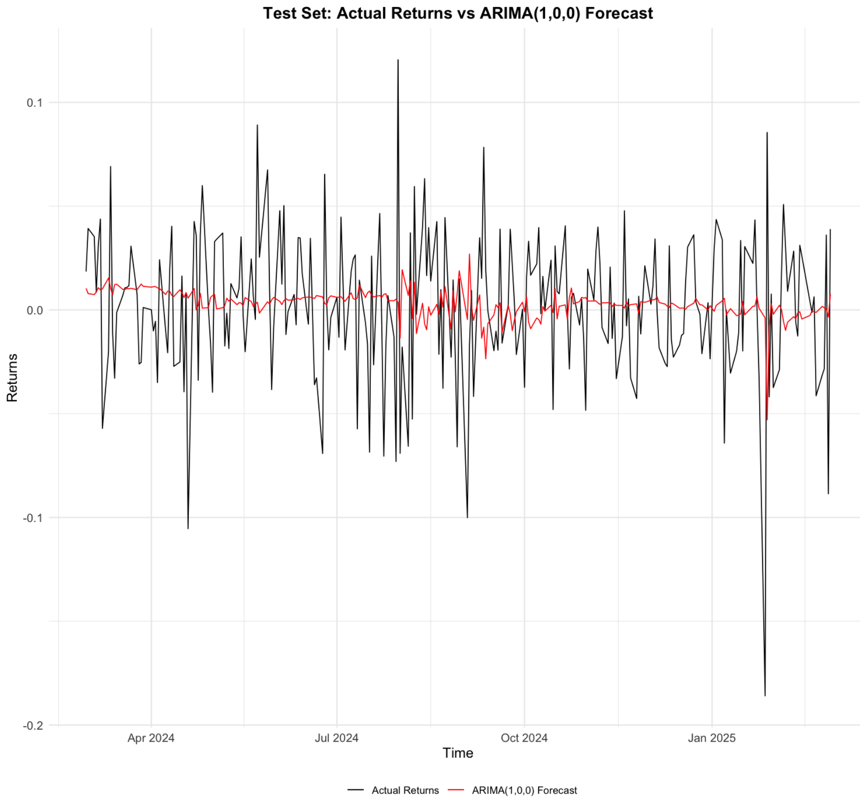


Figure 7: ARIMA(1,0,0) Forecasts on Test Set

Figure 7 illustrates the actual returns against the ARIMA(1,0,0) forecasts over the test period. The model clearly fails to capture much of the short-term variability, instead producing smoothed predictions that hover near the mean. This outcome is not unexpected. As established earlier, the return series displays near white noise behavior with minimal structure for a univariate linear model to exploit. Consequently, even though ARIMA(1,0,0) was the most effective among the candidates, its forecasts remain close to naïve predictions.

4.4 Hit Ratio Evaluation of ARIMA(1,0,0) Forecasts

To examine whether the ARIMA(1,0,0) model provides directional value for traders, we computed the hit ratio, defined as the proportion of days where the predicted sign of the return matches the actual sign. Out of 251 trading days in the test set, the model correctly predicted the return direction on 135 days, yielding a hit ratio of approximately 53.8%.

To assess whether this exceeds what could be expected from random guessing, we performed a binomial test under the null hypothesis of a 50% hit rate. The resulting p-value ($p > 0.10$) indicates that we cannot reject the null at conventional significance levels. Although the directional performance of the model is slightly above chance, this difference is not statistically significant. These results suggest that the ARIMA(1,0,0) model is insufficiently strong to support reliable directional trading based solely on one-day-ahead forecasts.

4.5 ARIMA Conclusion

These findings directly address our research question: Can ARIMA and GARCH models provide useful forecasts for NVIDIA's daily return and volatility? In the case of returns, the results suggest only limited forecasting power from ARIMA models. The series exhibits weak serial dependence and high unpredictability, which inherently constrains the potential of linear time-series models. Although ARIMA (1,0,0) provided slightly better performance than more complex alternatives, its practical utility remains poor in isolation.

The ARIMA model functions as intended by modeling the conditional mean of the returns. However, in daily financial data, this mean typically has little predictive content, which limits the relevance of the model for return forecasting. The next step is to evaluate whether GARCH models can add value by capturing the conditional volatility dynamics, something the ARIMA model, by design, is not suited to do.

4.6 GARCH Model Estimation on Training Set

To capture the time-varying volatility in NVIDIA's returns, we next estimate a GARCH model on the residuals from the ARIMA(1,0,0). Since the ARIMA(1,0,0) model accounts for the conditional mean, we apply a GARCH(1,1) model to its residuals, resulting in an ARIMA(1,0,0)-GARCH(1,1) specification to capture the time-varying volatility in NVIDIA's returns. We considered various GARCH(p,q) specifications and found that a GARCH(1,1) provides the best balance of fit and parsimony, outperforming higher-order alternatives according to BIC. This choice is further supported by empirical research showing that the GARCH(1,1) model consistently performs well in capturing volatility dynamics, making it a benchmark in volatility modelling (Portofolio Optimizer, 2024). Accordingly, we proceed with a GARCH(1,1), as the volatility model for NVIDIA's return series.

Parameter	**Estimate**	**Std. Error**	**t-Statistic**	**p-value**
μ	0.000279	0.000890	0.31305	0.7542
ω	0.000034	0.000016	2.08849	0.0368
α_1	0.078103	0.024144	3.23492	0.0012
β_1	0.892633	0.032225	27.7002	0.0000
Shape	6.719854	1.258600	5.33915	0.0000
RMSE (volatility)		0.0027		
MAE (volatility)		0.0012		

Table 4: In-Sample Estimation Results for GARCH(1,1) Model

The in-sample estimation of the GARCH(1,1) model reveals statistically significant effects of both recent shocks and persistent volatility. In particular, the ARCH coefficient ($\alpha_1 \approx 0.08$) is significant ($p < 0.01$), and the GARCH term ($\beta_1 \approx 0.89$) is also significant ($p < 0.01$). These estimates indicate that large return shocks contribute significantly to increases in next-day volatility (via the significant α_1 term), while the high β_1 value confirms strong persistence in volatility over time. Notably, the volatility is highly persistent: the sum $\alpha_1 + \beta_1 \approx 0.97$, implying that volatility shocks decay slowly over time (Stiglingh and Seitshiro, 2022).

As expected, the estimated intercept in the mean equation is effectively zero (and not statistically significant), as the mean dynamics are already captured by the AR(1) component. We also allow for heavy-tailed returns by using Student's t error distribution. The estimated degrees of freedom parameter is approximately 6.72. The in-sample forecasting performance proves to be reasonably good, with an RMSE of 0.0027 and MAE of 0.0012, indicating that the model tracks the conditional volatility with high precision during the training period.

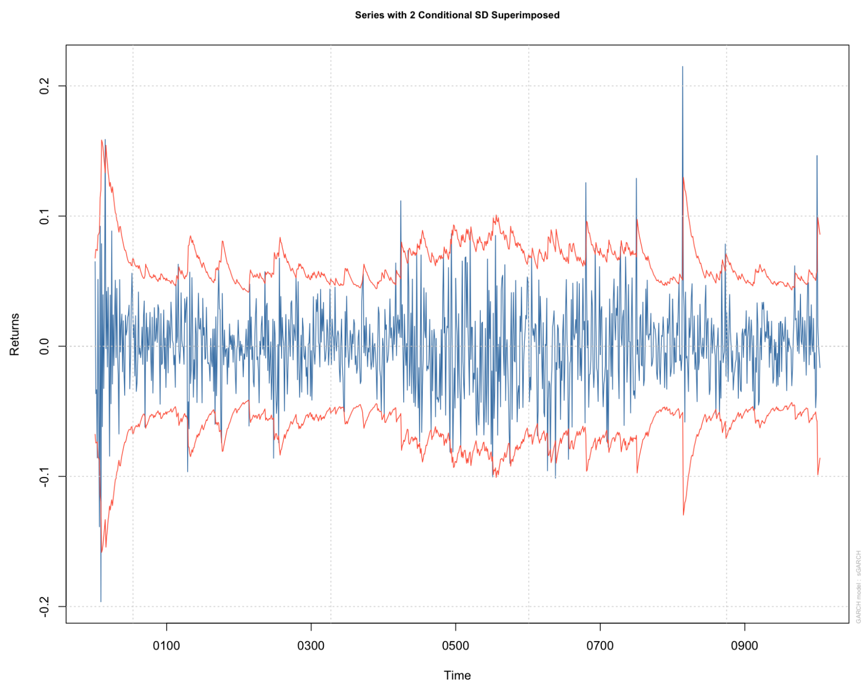


Figure 8: Returns with ± 2 Conditional Standard Deviations from GARCH(1,1) Model

Figure 8 displays the conditional standard deviation bands produced by the GARCH(1,1) model overlaid on the actual return series. The plot shows how the model dynamically adjusts volatility estimates in response to shocks, with wider bands during volatile periods and narrower bands during calmer periods. This visual evidence supports the model's ability to capture volatility clustering effectively.

Overall, the GARCH(1,1) fit captures the volatility clustering in the training set: past volatility and shocks strongly dictate the current volatility.

4.7 GARCH Final Forecasting and Evaluation

The GARCH(1,1) one-step-ahead forecasts achieve an out-of-sample **RMSE** of 0.0053 and **MAE** of 0.0042. These error magnitudes are very small in absolute terms, indicating that on average the predicted volatility is within roughly 0.42% of the realized volatility. In practical terms, if the true daily volatility is 2%, the model's forecast would typically fall within about $\pm 0.42\%$ of that value. This level of accuracy suggests that the GARCH(1,1) model effectively captures the general level of volatility.

Importantly, the slightly larger RMSE relative to the MAE indicates that the largest forecast errors tend to occur during abrupt volatility spikes or drops. In other words, the model generally tracks volatility well under normal conditions, but it tends to underpredict sudden jumps.

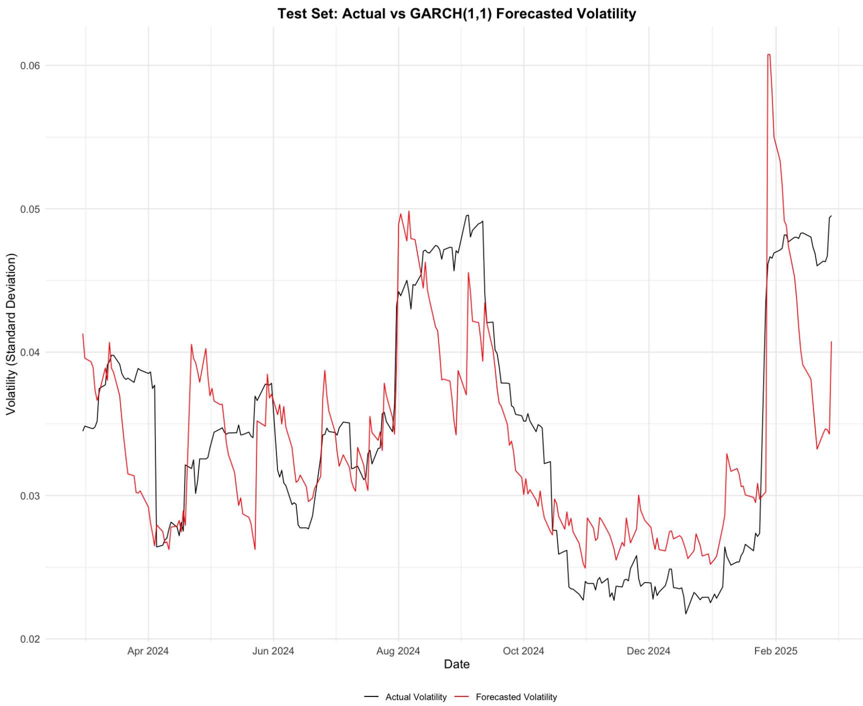


Figure 9: GARCH(1,1) Forecasts on Test Set

Figure 9 illustrates the out-of-sample volatility forecasting performance by plotting the GARCH(1,1) one-step-ahead predicted volatility against NVIDIA's realized 30-day volatility. Across the test period, the forecasted volatility captures the general level and direction of realized volatility reasonably well. The two series tend to move together, particularly during sustained periods of rising and falling volatility (volatility clustering), such as mid 2024 and late 2024. However, the GARCH(1,1) model struggles to respond to sudden spikes in volatility. For example, in early February 2025, NVIDIA's realized volatility increases abruptly, while the model's predicted volatility rises more slowly and fails to capture the magnitude of the jump. These observations highlight a weakness of the GARCH(1,1) framework: while it effectively models volatility clustering, it tends to smooth out sudden spikes, making it less responsive to sudden market turbulence. Overall, the results show that the GARCH(1,1) model offers a practically useful volatility forecast for NVIDIA in typical market environments, while highlighting the need for caution or enhancements during periods of abrupt change.

4.8 Post-Forecast Diagnostics

To ensure that our out-of-sample forecasts are well specified, we conducted diagnostic checks for both the ARIMA and GARCH models. For the ARIMA(1,0,0) model, we examined the forecast errors using ACF, PACF, and the Ljung–Box test. These tests showed no significant autocorrelation, suggesting the model performed adequately on the test data.

For the GARCH(1,1) model, we applied the Ljung–Box test to the standardized residuals, defined as residuals scaled by the model's forecasted volatility. Again, no significant autocorrelation was found. ACF and PACF plots further supported that the model captured the conditional variance effectively.

Together, these diagnostics indicate that both models are well specified in our forecasting: the ARIMA component captured the predictable structure in the mean, and the GARCH component handled the time-varying variance. The remaining residuals appear largely unpredictable and consistent with white noise, which is the desired outcome of a properly specified time series model.

5 Conclusion

This analysis of NVIDIA's stock confirms that short-term return predictions using ARIMA models remain extremely challenging. The ARIMA(1,0,0) model emerged as the best among those tested, but its daily return forecasts were only marginally better than a naive random-walk benchmark. Even when statistically significant, the ARIMA model's predictions provided limited practical value. This outcome aligns with the Efficient Market Hypothesis: prices quickly incorporate available information, so future daily returns largely reflect new, unpredictable events rather than past patterns (Fama, 1970).

By contrast, modeling volatility with a GARCH(1,1) model was much more productive. The GARCH model captured the well-known clustering of volatility in NVIDIA returns, high-volatility days tended to follow one another. Out-of-sample volatility forecasts showed reasonable accuracy and stability, even though the model smoothed over some abrupt volatility spikes. For investors and risk managers, this is key: volatility forecasts can inform risk assessments and position sizing in ways that return forecasts cannot.

We acknowledge several limitations of our approach. The analysis covers a five-year sample (2020–2025), which may not capture longer-term market shifts or structural changes. We also used only a single ARIMA specification and a GARCH(1,1) model. These models assume symmetric shock effects and may understate sharp market moves or "leverage" effects. Moreover, both models rely purely on historical prices and omit exogenous variables, potentially missing information from macroeconomic trends or related market indicators, especially during unusual conditions.

Future extensions could address these issues. More flexible volatility models, such as EGARCH or GJR-GARCH, allow for asymmetric responses to positive versus negative shocks and may better capture behavior during turbulent periods. Similarly, expanding the forecasting framework to include exogenous predictors (for example, through an Autoregressive Distributed Lag model) could enhance accuracy by incorporating market-wide or macroeconomic indicators. These improvements could refine the forecasts and provide more responsive tools in rapidly changing markets.

In summary, our findings are consistent with financial theory and empirical evidence. They reinforce the idea that, while stock returns are largely unpredictable in the short run, volatility exhibits persistence and structure that can be modeled to some extent. By highlighting this contrast, the project contributes to understanding financial forecasting: it underscores that models like ARIMA may have limited use for predicting returns, whereas volatility modeling with GARCH offers practical value for market participants managing risk.

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